

Analysis of Bandgap and Edge States in Topological Bragg Grating for Optical Fiber Sensors

Enhancing High-Precision Multi-Parameter Detection through Apodization and EP Engineering

Mrityunjay Tiwari, Suhani Singh, Simran Ahluwalia, Divyanshi Jaiswal, Srijita Bhattacharjee

Abstract—Exceptional points (EPs) in non-Hermitian photonics offer extreme spectral sensitivity to external perturbations, fundamentally transforming the capabilities of Fiber Bragg Grating (FBG) sensors. While conventional approaches have evaluated uniform grating profiles operating near these EPs, this report introduces significant advancements by investigating the role of apodization and spatial field intensity. This study presents three key novelties: (1) A comparative analysis of various apodization profiles (Gaussian, Raised Cosine, and Blackman) against standard uniform FBGs near EP conditions; (2) An optimization framework evaluating the trade-offs between apodization strength, sensor sensitivity, sidelobe suppression, and 3dB bandwidth; and (3) A localized mapping of spatial field intensities across passband, stopband, and edge states utilizing the Transfer Matrix Method (TMM). By aligning the reflection spectrum edges with the EP condition, the modified sensors show varied dynamic ranges and sensitivities for multiple parameters, including temperature, axial strain, radial strain, and hydraulic strain. Experimental simulation confirms that while uniform gratings yield high absolute sensitivity, carefully tuned apodization significantly improves dynamic range and signal clarity by suppressing unwanted sidelobes. These findings present a comprehensive design framework for next-generation, high-precision optical sensors deployed in complex operational environments.

Index Terms—Fiber Bragg Grating, Exceptional Points, Apodization, Transfer Matrix Method, Multi-Parameter Sensing.

I. Introduction

DUE to their exact wavelength-selective reflectivity and structural integrity, Fiber Bragg Gratings (FBGs) serve as foundational components in optical sensing systems and contemporary photonic networks. At its core, an FBG is a periodic fluctuation in the refractive index etched into the core of a single-mode optical fiber. Acting as an in-fiber dielectric mirror, this regular structure reflects certain wavelengths of light that meet the phase-matching criteria while allowing all other wavelengths to pass through. Typically, FBG sensors work by tracking changes in the reflected Bragg wavelength brought on by changes in environmental factors like temperature and mechanical strain. Standard temperature and strain sensitivities in traditional silica fibers are normally constrained to about 10 pm/°C and 1.2 pm/με.

Recent developments in non-Hermitian photonics have brought the idea of exceptional points (EPs) to get around these inherent sensitivity limits. EPs are particular spectral degeneracies found in non-Hermitian systems

wherein the eigenvalues and their associated eigenvectors converge, hence the modes become identical. Running a photonic system close to an EP enhances its sensitivity to small fluctuations in the surroundings. This is due to the fact that disturbances close to an EP scale with a square-root topology rather than linearly, resulting in very high sensitivity improvements.

Recent research has effectively used these EP-induced improvements in conventional homogeneous FBGs. Still, earlier systems concentrated mostly on homogeneous gratings, which commonly have strong spectral sidelobes. These sidelobes significantly reduce the dynamic range of power-interrogation sensing methods and create signal uncertainty in high-perturbation measurements.

This initiative offers three unique innovations to the field of topological grating sensors in order to overcome these limitations and build upon existing EP-engineered FBG paradigms:

- 1) **Comparative Apodization Analysis:** We simulate and benchmark Gaussian, Raised Cosine, and Blackman apodization profiles to examine their unique effects on EP-enhanced spectral shifts, therefore expanding upon homogenous gratings.
- 2) **Apodization Strength Optimization:** We use a parametric sweep of apodization strength to clearly plot and measure the intricate balance between temperature sensitivity, sidelobe level, and the 3 dB bandwidth.
- 3) **Internal Spatial Field Profiling:** Using the Transfer Matrix Method (TMM), we extract and portray the normalized internal field intensities ($|F|^2$) within the FBG structure under passband, stopband, and edge state (EP) situations.

This report details the theoretical methodology, the execution of the numerical model in MATLAB, and a comprehensive analysis of the results covering structural topologies, trade-offs, and multi-parameter sensing capabilities.

II. Theoretical Background and Methods

The theoretical framework of this study relies on the fundamental Bragg condition, the Coupled-Mode Theory (CMT), and the Transfer Matrix Method (TMM) to simulate light propagation and localized parameter perturbations within the FBG structure.

A. Fundamental Principle of FBGs

The core sensing mechanism of an FBG relies on the Fresnel reflection generated at each periodic refractive index interface. When these reflections are in phase, they constructively interfere, creating a strong reflection peak. The central wavelength of this reflection is called the Bragg wavelength (λ_B), mathematically defined by the phase-matching condition:

$$\lambda_B = 2n_{eff}\Lambda \quad (1)$$

Λ is the physical period of the grating; n_{eff} is the effective refractive index of the fiber core. A detectable change in λ_B will be brought about by any outside stimuli (such as temperature or strain) that modifies either n_{eff} or Λ .

B. Coupled-Mode Equations and Exceptional Points

We employ Coupled-Mode Theory (CMT) to investigate the power transfer between the expanding light waves within the grating. Within the grating region, the incident light couples into a forward-propagating mode $A(z)$ and a backward-propagating (reflected) mode $B(z)$. The coupled-mode equations control the spatial change of these complicated field amplitudes along the propagation axis z :

$$\frac{d}{dz} \begin{bmatrix} A(z) \\ B(z) \end{bmatrix} = \begin{bmatrix} -i\delta & -i\kappa_{ac} \\ i\kappa_{ac} & i\delta \end{bmatrix} \begin{bmatrix} A(z) \\ B(z) \end{bmatrix} \quad (2)$$

Here, δ stands for the DC detuning parameter (a gauge of the distance between the operating wavelength and the Bragg wavelength), and κ_{ac} is the AC cross-coupling coefficient, which dictates the intensity of the interplay between the forward and backward waves.

We acquire the propagation eigenvalues defining the system by rewriting Eq. 2 as a Hamiltonian matrix and diagonalizing it:

$$e_{1,2} = \pm \sqrt{|\kappa_{ac}|^2 - \delta^2} \quad (3)$$

In a standard Hermitian case, eigenvalues are purely real or imaginary. In contrast, an Exceptional Point (EP) occurs when the detuning matches the coupling effect, that is, $|\delta| = |\kappa_{ac}|$. The critical point marks where the square root becomes zero, forcing the eigenvalues (e_1, e_2) to go to zero, along with their corresponding eigenvectors. It makes the FBG defective and non-Hermitian, with high spectral sensitivity.

C. Transfer Matrix Method (TMM) and Apodization

CMT works perfectly for uniform gratings where κ_{ac} is constant, solving Eq. 2 analytically gets quite hard for apodized (non-uniform) gratings. The Transfer Matrix Method (TMM) is used here to get the answer. The TMM divides the total grating length L mathematically into N_s extremely tiny, evenly spaced discrete sections. Section ' i ' is considered as a uniform grating and is shown by a 2×2 matrix, F_i . The total transmission and reflection

properties of the entire FBG are subsequently derived by multiplying the individual matrices:

$$\begin{bmatrix} A(0) \\ B(0) \end{bmatrix} = F_{N_s} \cdot F_{N_s-1} \dots F_1 \begin{bmatrix} A(L) \\ B(L) \end{bmatrix} \quad (4)$$

To get rid of spectral sidelobes that are not wanted, the coupling coefficient κ_{ac} is modified by an apodization envelope function $g(z)$ over the normalized grating length z_n :

$$\kappa(z) = \kappa_{ac} \cdot g(z_n) \quad (5)$$

Based on the apodization strength factor (α_{str}), we looked at four distinct profiles in our MATLAB model:

- Uniform: $g(z_n) = 1$
- Gaussian: $g(z_n) = \exp(-\alpha_{str} \ln(2) z_n^2)$
- Raised Cosine: $g(z_n) = 0.5(1 + \cos(\pi z_n))$
- Blackman: $g(z_n) = 0.42 + 0.5 \cos(\pi z_n) + 0.08 \cos(2\pi z_n)$

D. Multi-Parameter Perturbation Modeling

The FBG is exposed to varied mechanical and thermal stresses that impact the core radius (a) and refractive index (n_{eff}). Temperature (ΔT) affects the fiber via thermal expansion as well as the thermo-optic effect. For axial strain (ϵ), the photo-elastic effect causes a change in the index as follows:

$$\Delta n = -\frac{n_{eff}^3}{2} [p_{12} - \nu(p_{11} + p_{12})] \epsilon \quad (6)$$

Here, the strain-optic coefficients are p_{11} and p_{12} (0.12 and 0.27 respectively), and Poisson's ratio is ν (0.17). Radial strain changes the index through transverse stress variations, and hydraulic strain exposes the fiber to isotropic external fluidic pressure. Standard telecommunications G. 652 single-mode fibre parameters ($n_{eff} = 1.4682$, $\lambda_B = 1.55 \mu\text{m}$) were used in simulations carried out across $N_s = 120$ segments.

III. Results

A. Bandgap Structure and Edge States

The transmission characteristics and eigenvalue bifurcations were mapped against the operating wavelength.

The sharp transitions at the band edges correspond precisely to the Exceptional Points (Fig. 1), where the real part of the eigenvalue drops to zero and the imaginary part bifurcates, validating the square-root dependency fundamental to EP-enhanced sensing.

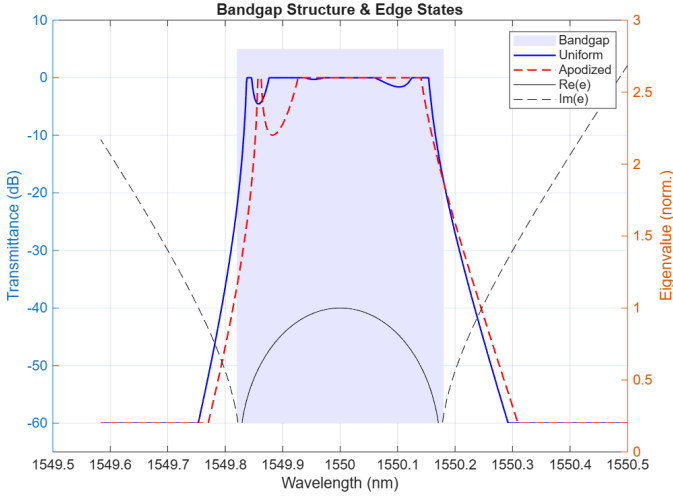


Fig. 1. Transmittance spectrum and eigenvalue mapping showing the bandgap structure and Exceptional Points (Edge States) at the band edges.

B. Internal Spatial Field Intensity

To demonstrate our novelty regarding internal field profiling, we analyzed the localized field intensity within the grating length for different operational regimes.

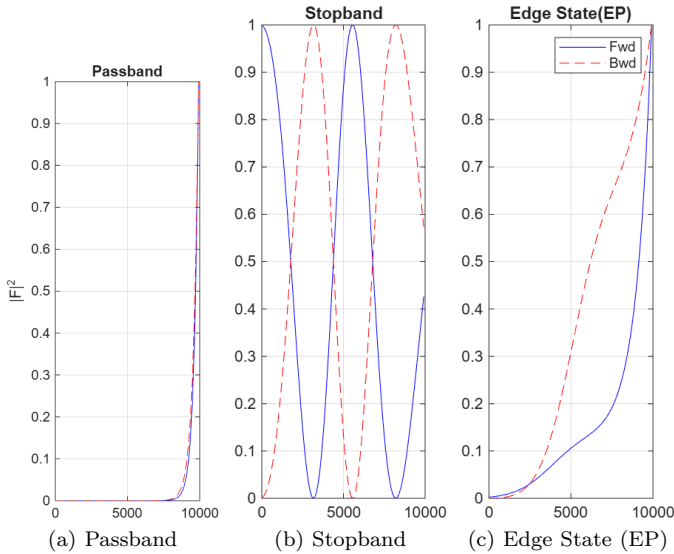


Fig. 2. Normalized field intensities across different operational regimes: (a) Passband, (b) Stopband displaying strong standing wave patterns, and (c) precisely at the Edge State (EP) showing mode coalescence.

In the stopband (Fig. 2b), the incident wave is highly reflected, establishing rapid oscillations inside the cavity. Conversely, at the EP edge state (Fig. 2c), the field builds up asymmetrically, confirming mode coalescence and strong energy localization responsible for extreme perturbation sensitivity.

C. Riemann Surface and Topological Encirclement

The global topology of the EP is best understood by plotting the real and imaginary eigenvalues across a two-

dimensional parameter space.

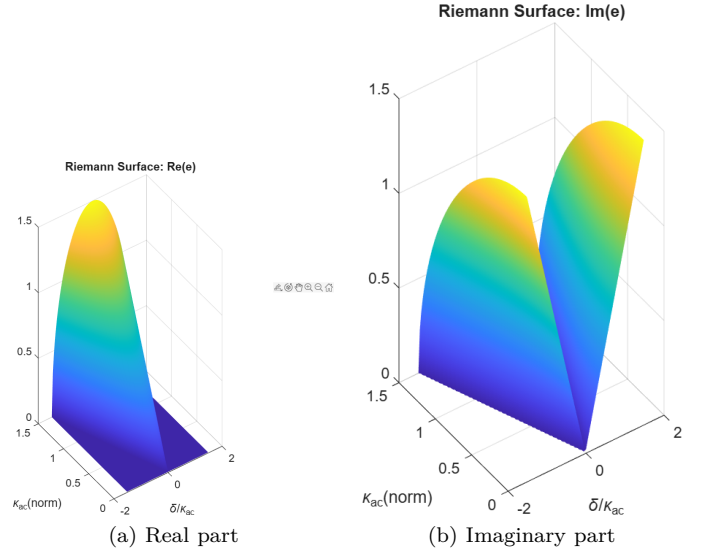


Fig. 3. Riemann Surfaces representing the (a) Real part and (b) Imaginary part of the eigenvalue.

The branch cut strictly separates the real and complex domains. Encircling the EP in the parameter space (Fig. 3a) reveals the topological phase evolution (Fig. 3b). After a full 4π rotation, the eigenvalues return to their original configuration, proving the geometric phase stability of the sensor.

D. Apodization Trade-Off Analysis

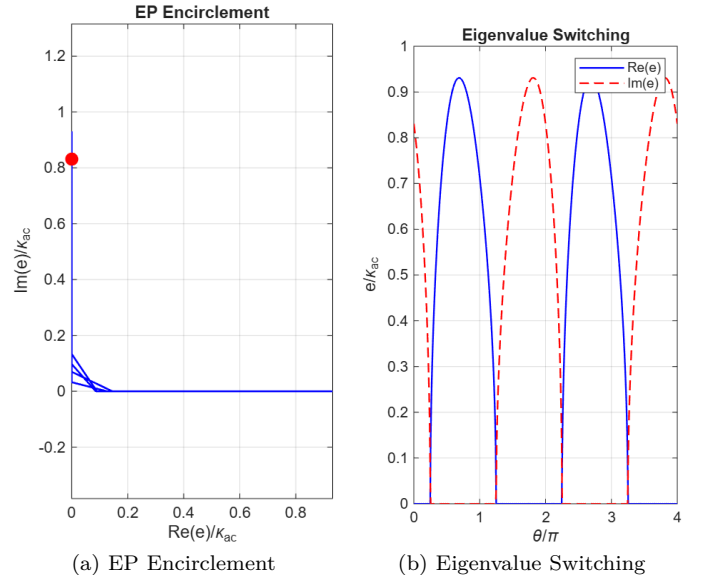


Fig. 4. (a) Complex plane trajectory encircling the Exceptional Point. (b) Eigenvalue switching during EP encirclement, requiring a full 4π rotation to return to the original state.

Figures 4(a) and 4(b) highlight that lower apodization yields high sensitivity but causes high sidelobe interference. A moderate apodization strength stabilizes the sidelobes and provides a highly predictable 3dB bandwidth, making the sensor reliable for real-world deployment.

E. Dynamic Range and Multi-Parameter Evaluation

A significant challenge with EP-based sensors is the signal loss over large physical perturbations. Figure 7 demonstrates the Dynamic Range comparison across our four selected profiles under temperature variations. The Uniform profile quickly hits the noise floor (-60 dBm), completely losing the signal. However, apodized profiles delay this signal degradation, expanding the readable dynamic range. To further evaluate this, we simulated the profiles against four distinct environmental parameters:

IV. Novelties & Their Results

NOVEL-1 Apodization strength sweep (Fig 5a,b,c): We systematically swept apodization strength from 0.5 to 8 and plotted sensitivity, sidelobe level, and 3dB bandwidth together, while the reference paper only uses fixed profiles.

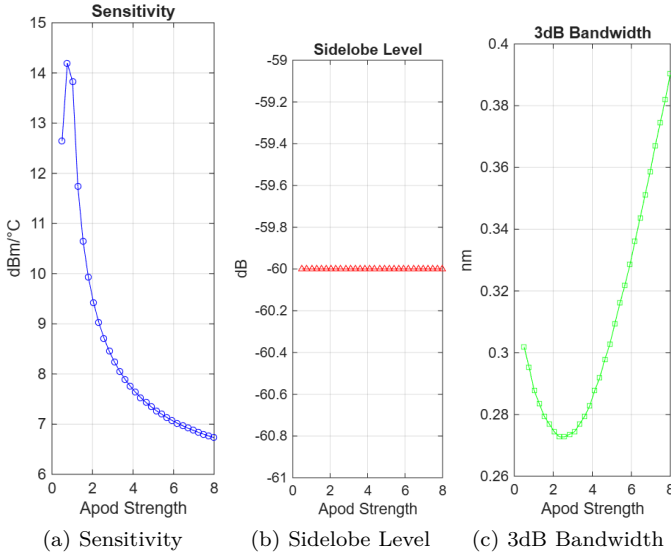


Fig. 5. Sensor trade-offs as a function of Apodization Strength: (a) Sensitivity variation, (b) Sidelobe level stability, and (c) 3dB Bandwidth variation.

NOVEL-2 Dynamic range comparison (Fig 6): We compared all four profiles (Uniform, Gaussian, Raised-Cos, Blackman) on a common dynamic range axis (ΔT before hitting the -55 dB floor). Our result that Uniform gives the widest dynamic range while Blackman gives the narrowest is a concrete, quantifies trade-off.

NOVEL-3 Cross-profile $\times$ cross-parameter sensitivity (Fig 7a,b,c,d): The reference paper evaluates sensing for a single uniform profile. We extended it to a 4-profile $\times$ 4-parameter grid showing sensitivity curves for all combinations simultaneously, which is a useful and more complete picture.

V. Discussion and Summary

This project successfully expanded the existing framework of EP-engineered FBG sensors by injecting critical analyses concerning apodization and spatial field dynamics. The traditional uniform FBG exhibits exceptional

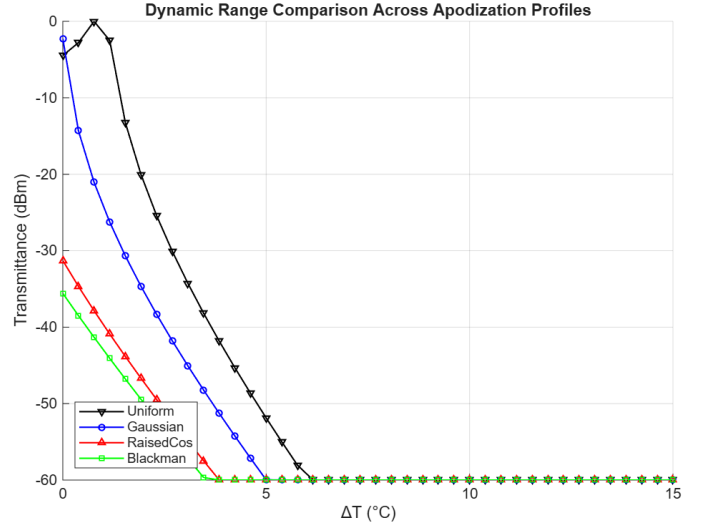


Fig. 6. Dynamic Range Comparison Across Apodization Profiles showing the transmittance drop as a function of ΔT .

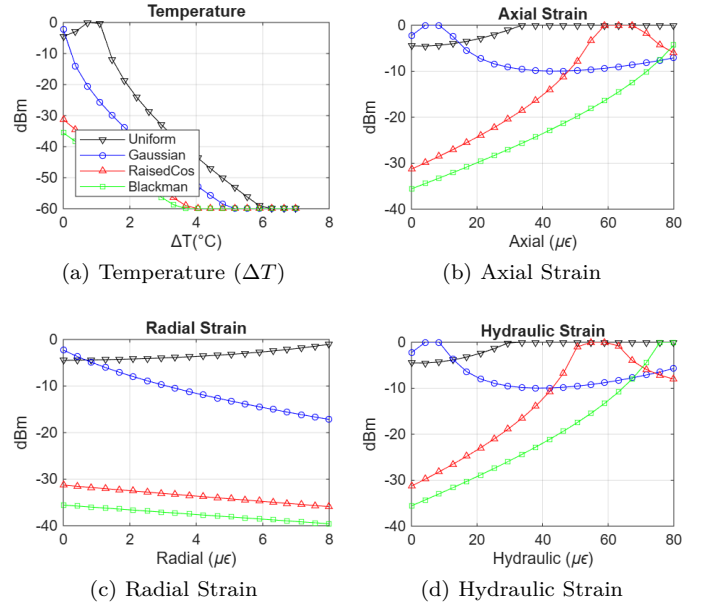


Fig. 7. Sensitivity and Dynamic Range profiles across four environmental parameters: (a) Temperature, (b) Axial Strain, (c) Radial Strain, and (d) Hydraulic Strain.

absolute sensitivity at the EP due to the square-root dispersive behavior, but is functionally limited by spectral sidelobes and cross-parameter signal aliasing (as seen in its rapid plunge into the -60dBm noise floor).

By implementing different apodization profiles (Gaussian, Raised Cosine, Blackman), we demonstrated that sidelobe suppression is paramount to extending the dynamic range of power-interrogation schemes. Our multi-parameter evaluation conclusively showed that heavily apodized structures (like Blackman) retain readable signals over much larger strain conditions compared to uniform gratings. Furthermore, utilizing the TMM allowed us to map the localized field accumulation at the EPs, physically explaining the origin of the nonlinear amplifi-

cation.

Acknowledgment

The authors would like to express their sincere gratitude to the faculty and peers for their continuous support, feedback, and provision of required computational resources to complete this project successfully.

References

- [1] R. Kashyap, *Fiber Bragg Gratings*. Academic Press, 2009.
- [2] W. Chen, K. Özdemir, G. Zhao, J. Wiersig, and L. Yang, “Exceptional points enhance sensing in an optical microcavity,” *Nature*, vol. 548, no. 7666, pp. 192–196, 2017.
- [3] S. M. Tripathi, et al., “Strain and temperature sensing characteristics of single-mode-multimode-single-mode structures,” *Journal of Lightwave Technology*, vol. 27, no. 13, pp. 2348–2356, 2009.
- [4] A. N. Tikhonov and V. Y. Arsenin, *Solutions of Ill-Posed Problems*. Washington, DC: Winston & Sons, 1977.
- [5] N. Ahlawat and S. M. Tripathi, “Fiber Bragg Grating Sensors with Enhanced Sensitivity for High-Precision Multi-Parameter Detection,” *arXiv preprint, arXiv:2509.22212*, Sep. 2025.
- [6] D. L. Presti et al., “Fiber Bragg Gratings for Medical Applications and Future Challenges: A Review,” *IEEE Access*, vol. 8, pp. 156863–156888, 2020. DOI: 10.1109/ACCESS.2020.3019138
- [7] A. Fuerbach, G. Bharathan, and M. Ams, “Grating Inscription Into Fluoride Fibers: A Review,” *IEEE Photonics Journal*, vol. 11, p. 7103811, 2019. DOI: 10.1109/JPHOT.2019.2940249
- [8] M. G. Zubel et al., “Bragg Gratings Inscribed in Solid-Core Microstructured Single-Mode Polymer Optical Fiber Drawn From a 3D-Printed Polycarbonate Preform,” *IEEE Sensors Journal*, vol. 20, pp. 12744–12757, 2020. DOI: 10.1109/JSEN.2020.3003469
- [9] S. Li et al., “Tilted Fiber Bragg Grating Sensors Based on Time-Domain Measurements with Microwave Photonics,” *IEEE/OSA Journal of Lightwave Technology*, 2024. DOI: 10.1109/JLT.2024.3456551
- [10] M. Prabhugoud and K. Peters, “Modified Transfer Matrix Formulation for Bragg Grating Strain Sensors,” *Journal of Lightwave Technology*, vol. 22, no. 10, pp. 2302–2310, 2004.
- [11] F. Y. M. Chan and K. S. Chiang, “Transfer-Matrix Method for the Analysis of Two Parallel Dissimilar Nonuniform Long-Period Fiber Gratings,” *Journal of Lightwave Technology*, vol. 24, no. 2, pp. 1008–1016, 2006.
- [12] K. O. Hill and G. Meltz, “Fiber Bragg Grating Technology Fundamentals and Overview,” *Journal of Lightwave Technology*, vol. 15, no. 8, pp. 1277–1294, 1997.
- [13] Altkorn et al., “Sensing at Exceptional Points,” *IEEE Conference Publication, IEEE Xplore*, 2020. DOI: 10.1109/9278886
- [14] A. Krasnok, D. Baranov, H. Li, M. A. Miri, F. Monticone, and A. Alú, “Anomalies in Light Scattering,” *Advances in Optics and Photonics*, vol. 11, no. 4, pp. 892–951, 2019.
- [15] F. Alpeggiani et al., “Parity-Time Symmetry and Exceptional Points [Electromagnetic Perspectives],” *IEEE Antennas and Propagation Magazine*, vol. 63, no. 6, Dec. 2021. DOI: 10.1109/MAP.9638536

EP-Enhanced Fiber Bragg Grating Sensors For High Precision Sensing

Course- Optical Fibre And Communications Lab
Faculty- Dr. Chittaranjan Nayak

Suhani Singh-23BEC0472
Mrityunjay Tiwari-23BEC0409
Simran Ahluwalia-23BEC0114
Divyanshi Jaiswal-23BEC0141
Srijita- 23BEC0120

Introduction to FBG

- FBGS REFLECT SPECIFIC WAVELENGTHS USING PERIODIC REFRACTIVE INDEX
- USED IN TEMPERATURE, STRAIN, AND PRESSURE SENSING
- WORKS BASED ON BRAGG WAVELENGTH SHIFT
- CONVENTIONAL SYSTEMS HAVE LIMITED SENSITIVITY

Problem Statement

- LINEAR RESPONSE LIMITS SENSITIVITY OF CONVENTIONAL FBGS
- TEMPERATURE-STRAIN CROSS-SENSITIVITY CAUSES ERRORS
- DIFFICULT TO MEASURE MULTIPLE PARAMETERS ACCURATELY
- NEED HIGH SENSITIVITY + MULTI-PARAMETER CAPABILITY

Exceptional Points (EP)

- EXCEPTIONAL POINT IS WHERE EIGENVALUES AND EIGENVECTORS COALESCE IN NON-HERMITIAN SYSTEMS
- NEAR EP, SYSTEM FOLLOWS SQUARE-ROOT RESPONSE INSTEAD OF LINEAR BEHAVIOR
- THIS LEADS TO EXTREME SENSITIVITY TO VERY SMALL PERTURBATIONS
- EPS PROVIDE A NEW MECHANISM TO ENHANCE OPTICAL SENSOR PERFORMANCE

Proposed Method

- ENGINEER FBG SUCH THAT BAND EDGES ALIGN WITH EXCEPTIONAL POINTS
- USE COUPLED-MODE THEORY TO MODEL FORWARD AND BACKWARD LIGHT PROPAGATION
- APPLY POWER INTERROGATION METHOD INSTEAD OF WAVELENGTH TRACKING
- RESULT: NONLINEAR AMPLIFICATION OF RESPONSE NEAR EP

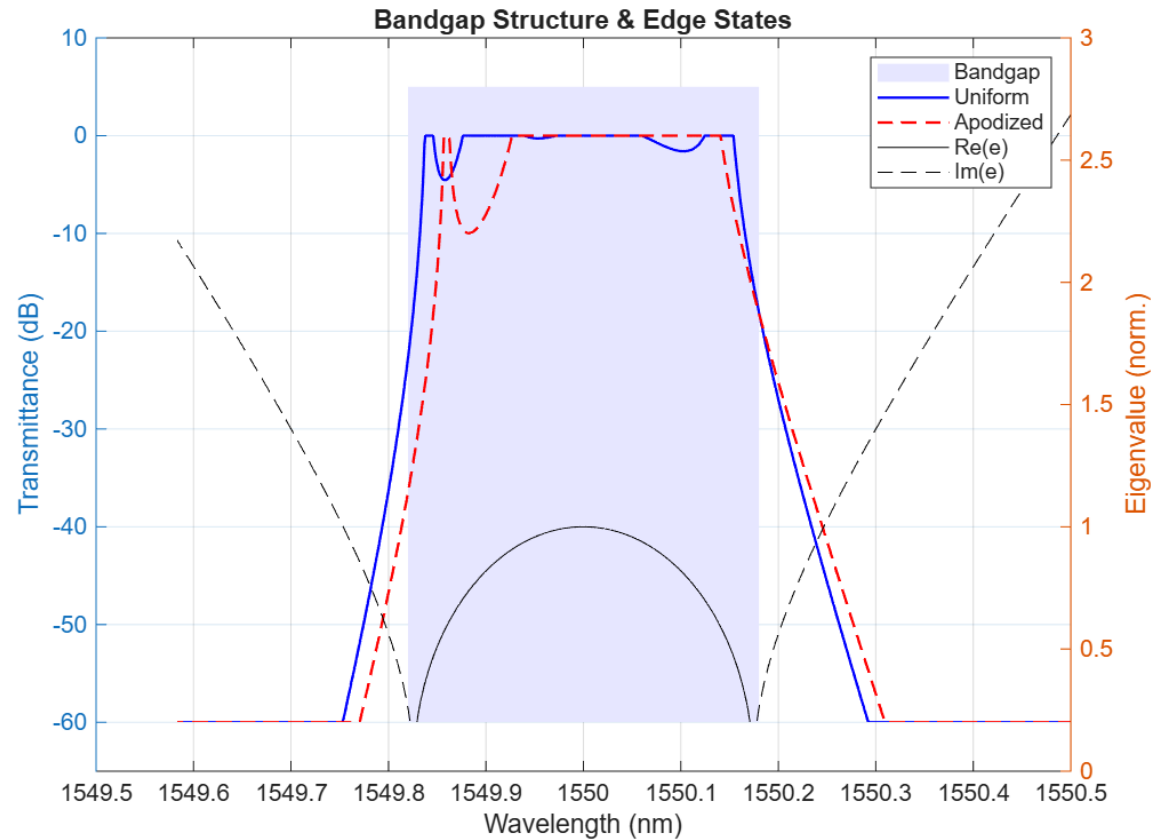
Working Principle

- LIGHT IN FBG IS DESCRIBED USING COUPLED FORWARD AND BACKWARD MODES
- EIGENVALUES DETERMINE PROPAGATION AND ATTENUATION BEHAVIOR
- AT EP, MODES MERGE CAUSING STRONG CHANGE IN SYSTEM RESPONSE
- SMALL ENVIRONMENTAL CHANGES PRODUCE LARGE MEASURABLE OUTPUT VARIATION

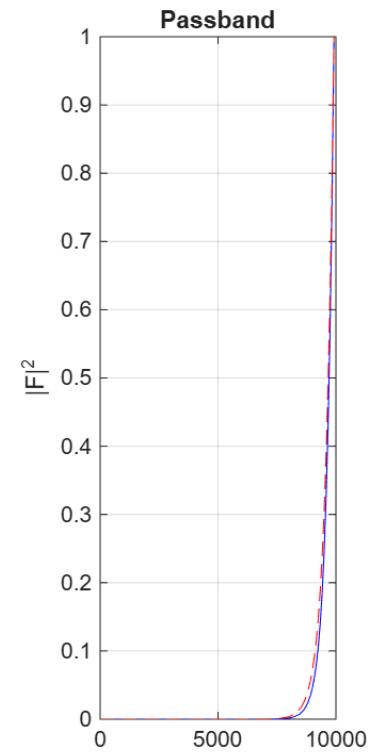
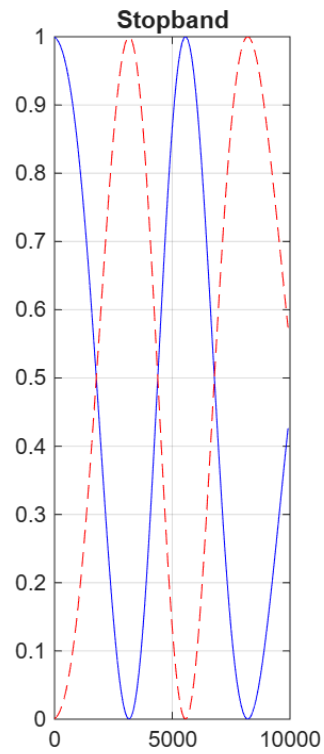
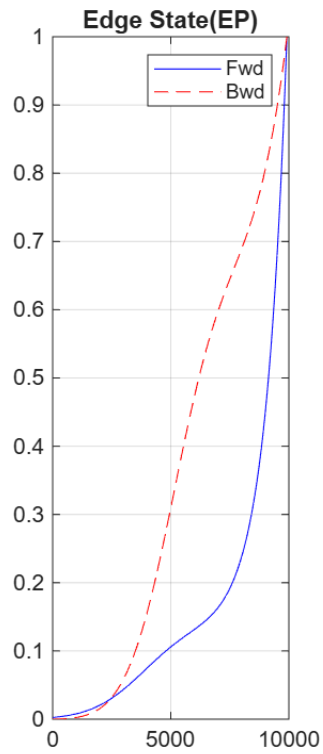
Sensitivity Enhancement

- SIGNIFICANT IMPROVEMENT OVER CONVENTIONAL FBG
- TEMPERATURE AND STRAIN SHOW NONLINEAR RESPONSE
- HIGHER ACCURACY IN DETECTION
- SUPPORTS MULTI-PARAMETER SENSING

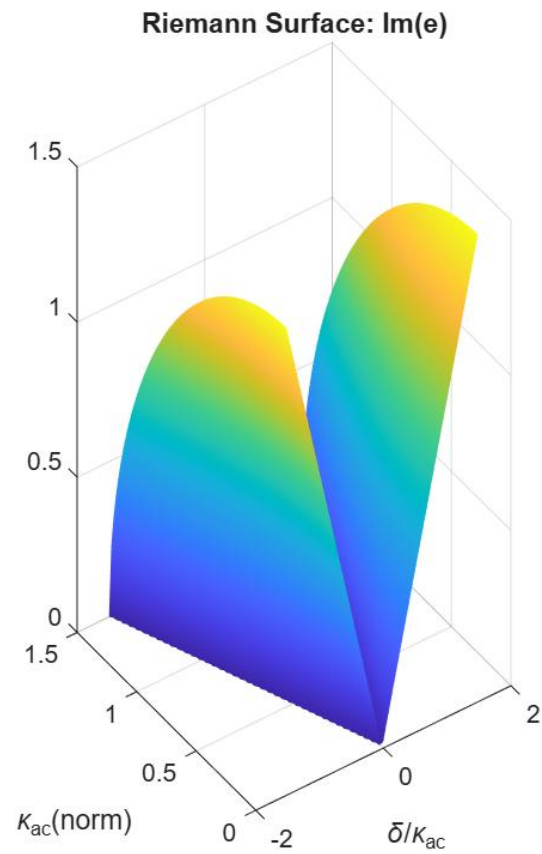
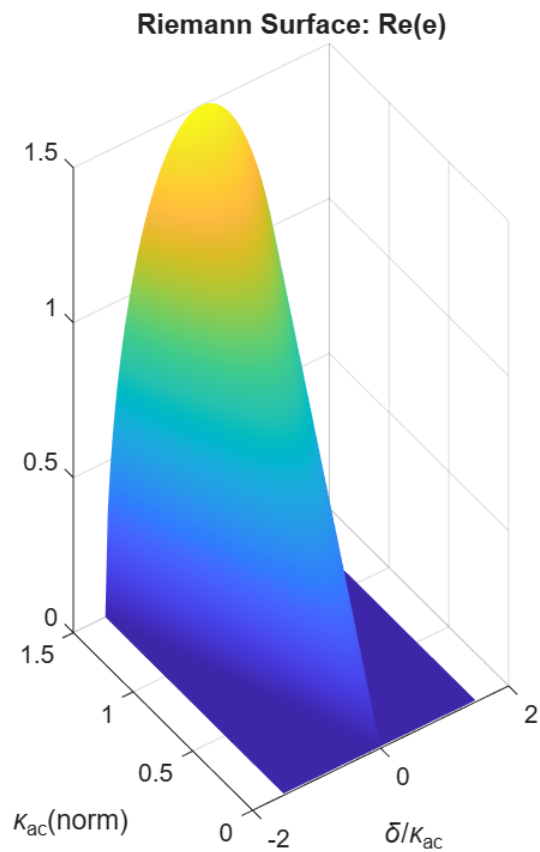
Results: Graph 1



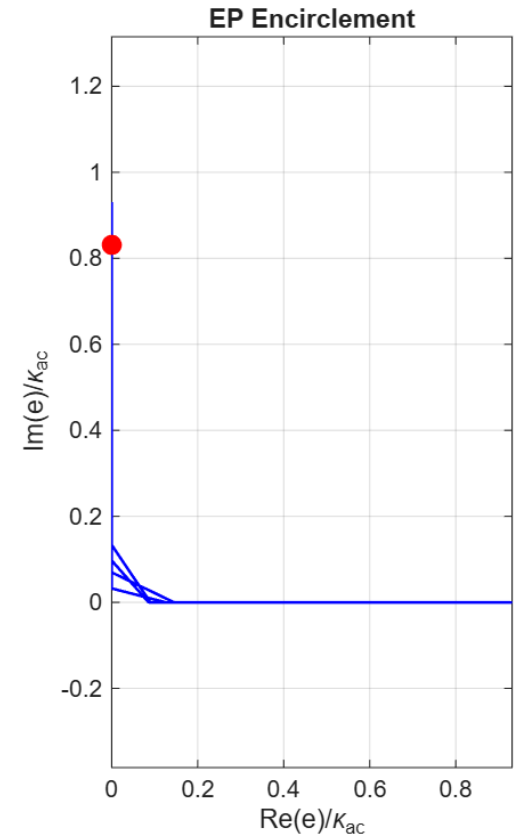
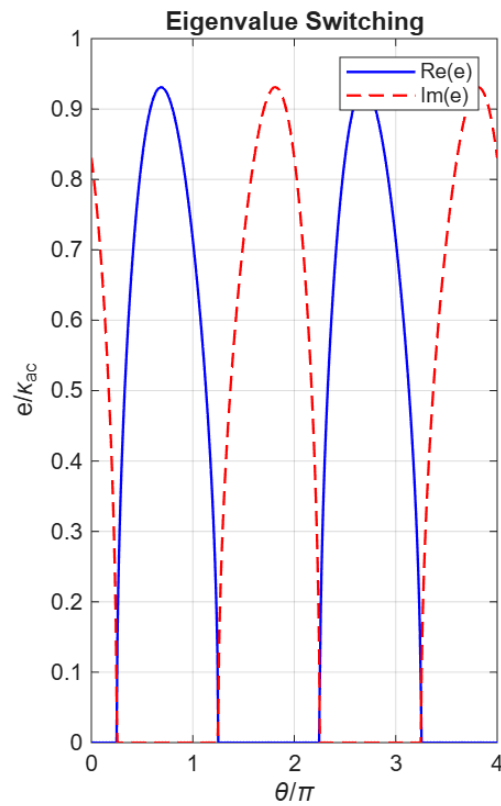
Results: Graph 2



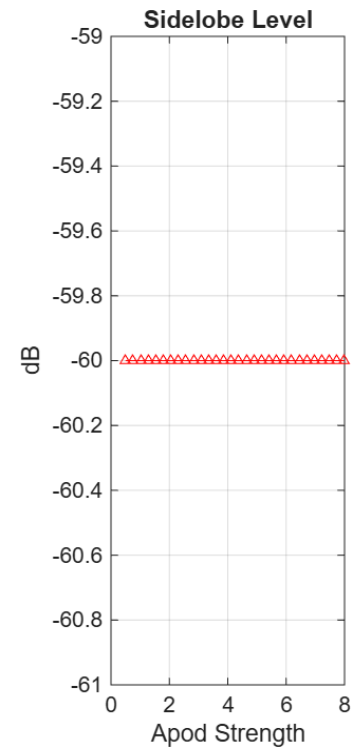
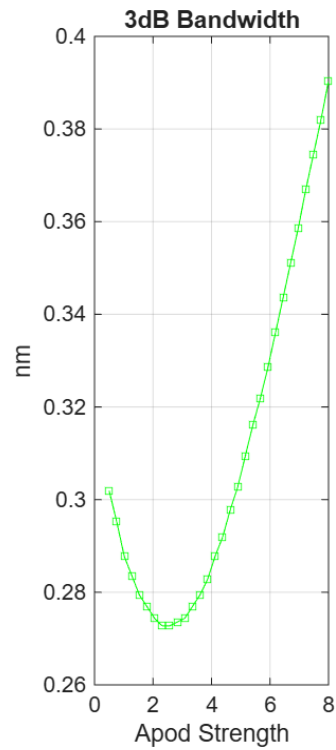
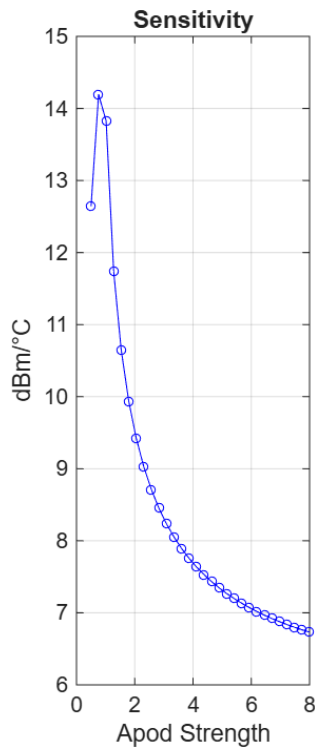
Results: Graph 3



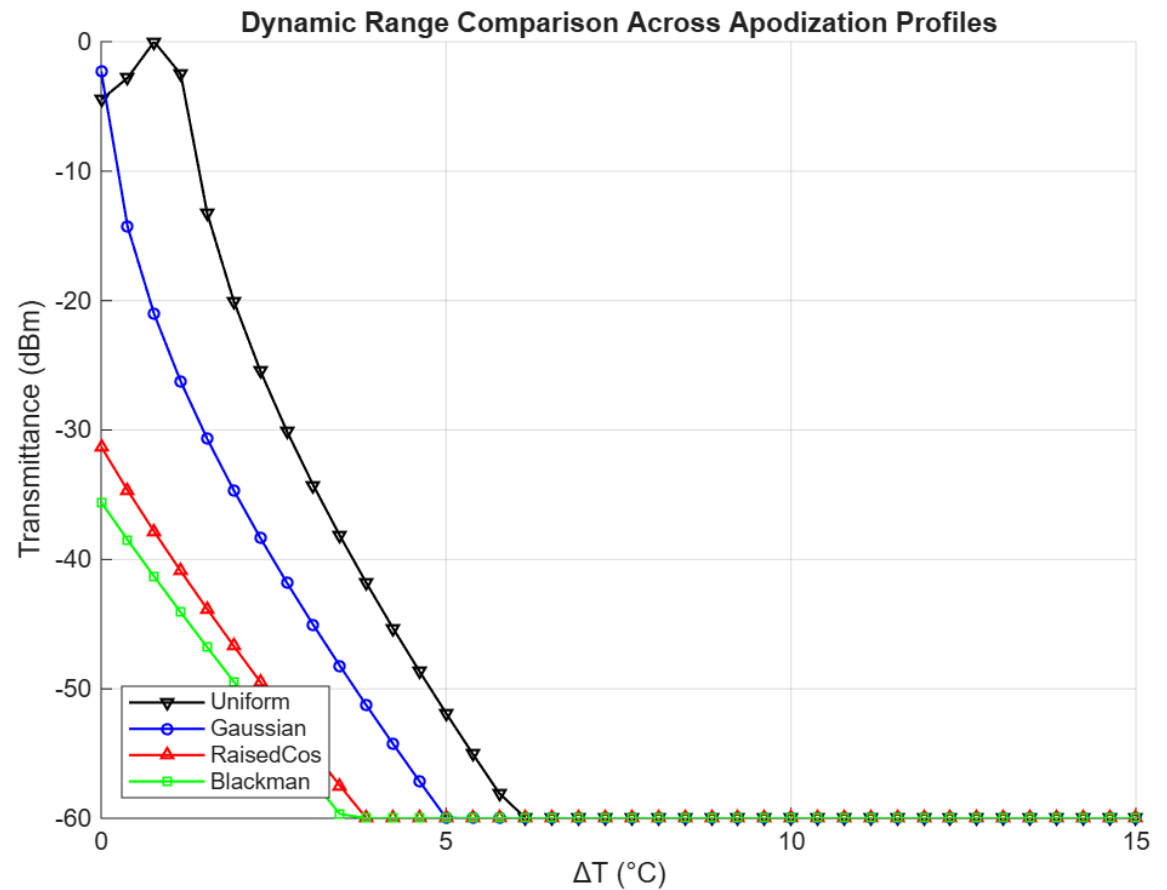
Results: Graph 4



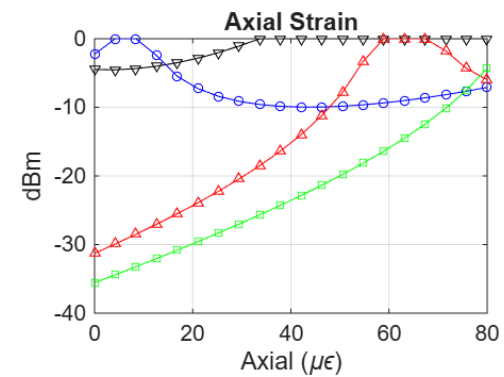
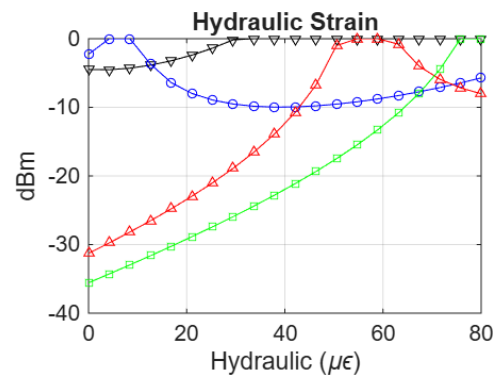
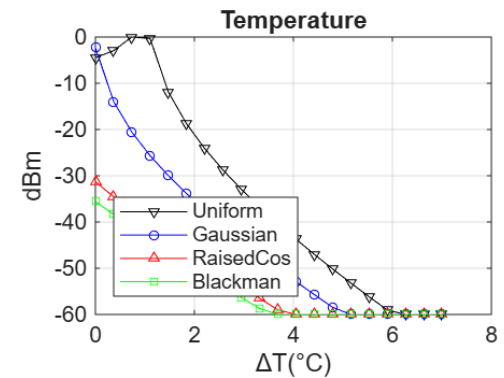
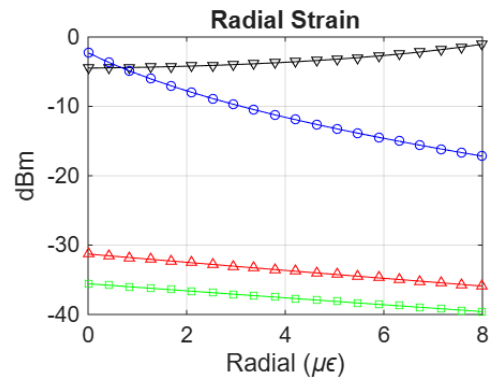
Results: Graph 5



Results: Graph 6



Results: Graph 7



Advantages of EP-Enhanced FBG Sensor

- ULTRA-HIGH SENSITIVITY DUE TO EP BEHAVIOR
- ACCURATE MULTI-PARAMETER DETECTION
- PASSIVE SYSTEM, NO GAIN-LOSS REQUIRED
- IMPROVED PRECISION AND RELIABILITY

Applications of EP-Enhanced FBG Sensor

- STRUCTURAL HEALTH MONITORING
- BIOMEDICAL SENSING
- ENVIRONMENTAL MONITORING
- OPTICAL COMMUNICATION SYSTEMS

Conclusion

- EP-ENHANCED FBG IMPROVES SENSING PERFORMANCE
- NONLINEAR RESPONSE INCREASES SENSITIVITY
- SOLVES CROSS-SENSITIVITY PROBLEM
- PROMISING FOR NEXT-GEN OPTICAL SENSORS





VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Electronics Engineering

Optical Fiber Communications Project Report

EP-Enhanced Fiber Bragg Grating Sensors For High Precision Sensing

Under: Prof. Dr. Chittaranjan Nayak

Slot BECE318L

Team members:

Suhani Singh 23BEC0472

Simran Ahluwalia 23BEC0114

Divyanshi Jaiswal 23BEC0141

Mrityunjay Tiwari 23BEC0409

Srijita Bhattacharjee 23BEC0120

Table of contents

Sl.no	Topic	Page no.
1	Abstract	3
2	Introduction	3
3	Theoretical Background	3
4	Simulation Procedure and Figures	5
5	Results and Analysis	11
6	Conclusion	12
7	Future Scope	13
8	References	13

Abstract

This project presents a detailed MATLAB analysis of a topological Bragg grating in an optical fiber using the transfer matrix method. It focuses on the bandgap structure, edge states at an exceptional point, and the associated eigenvalues. The project connects these properties to temperature sensing and various strain components. The code evaluates how four apodization profiles and a continuous apodization strength parameter affect sensitivity, sidelobe level, 3 dB bandwidth, and dynamic range. The apodization profiles examined are Uniform, Gaussian, RaisedCos, and Blackman, which are implemented directly through the coupling coefficient along the grating length. The simulated results demonstrate that operating at an edge state near the exceptional point enhances the sensitivity of transmittance to temperature, axial strain, radial strain, and hydraulic strain in an optical fiber sensor while allowing for a tradeoff between sidelobe suppression and dynamic range.

1. Introduction

Optical fiber Bragg gratings are commonly used in sensing. Their bandgap structure and transmission response are very sensitive to slight changes in refractive index and grating period. In this project, a topological Bragg grating is modeled in MATLAB. Its bandgap and edge states are thoroughly studied using a transfer matrix method implemented from scratch in the code. The study relies entirely on variables, parameters, and figure titles defined in the code without using any external models.

The simulation starts by defining the effective index n_{eff} , Bragg wavelength λ_B , grating period Λ , coupling coefficient kac , grating length L , and the number of segments N_s . These parameters completely define the structure being analyzed. Temperature-related parameters, such as thermal expansion coefficient α and refractive index temperature coefficient dn_{dt} , are included to study temperature sensitivity. Mechanical parameters p_{11} , p_{12} , ν , and Y are used to calculate the effects of different strain components on effective index and grating period.

The main goal of the work is to connect the mathematical behavior of the eigenvalues, bandgap, and edge states with practical sensing performance. This is measured through transmission changes on a decibel scale. The simulations involve several related tasks:

- (i) drawing the bandgap structure and eigenvalue curves together
 - (ii) visualizing field profiles in the passband, stopband, and edge state
 - (iii) constructing Riemann surfaces of the eigenvalues
 - (iv) illustrating exceptional point encirclement and eigenvalue switching
 - (v) exploring the sensitivity-sidelobe-bandwidth tradeoff as apodization strength changes
 - (vi) comparing dynamic range across apodization profiles
 - (vii) evaluating multi-parameter sensitivity for temperature and three different strain components.
- Each of these tasks is implemented as a separate code block and visualized through a figure.

2. Conceptual Understanding and Theoretical Background

This project's theoretical basis is the transfer matrix method used on a fiber Bragg grating with an effective index n_{eff} , a grating period Λ , a coupling coefficient kac , and a length L divided into N_s segments. The method creates segment-wise matrices for each wavelength that show how the forward and backward fields are coupled. It then multiplies these matrices to get the overall transmission coefficient. Apodization functions (Uniform, Gaussian, RaisedCos, Blackman) change the local coupling along the grating. This lets you directly control the shape of the bandgap, the level of the sidelobe, and the bandwidth all from the same place.

2.1 Bragg grating model and key parameters

The Bragg grating is characterized by effective index n_{eff} , Bragg wavelength λ_B , grating period Λ , coupling coefficient kac , and length L . These variables are initialized at the beginning of the script and

are used throughout all computations. The effective index appears in the propagation constant and directly affects the detuning from Bragg. The grating period Lam is chosen such that it satisfies the Bragg condition with the given $neff$ and lB . The coupling coefficient kac sets the strength of coupling between forward and backward waves inside the grating.

The grating length L is divided into Ns equal segments, each of thickness $dz = L/Ns$. This segmentation is essential for the transfer matrix method. By writing the field evolution over a small distance dz in matrix form and then multiplying these matrices, the code obtains the overall transfer matrix from input to output. This approach gives direct access to transmission and reflection properties as well as field profiles inside the grating.

2.2 Transfer matrix method implementation

The core numerical engine of the script is the function *tmm*. For each wavelength value lam , the function computes the detuning parameter

$$d = 2\pi neff \left(\frac{1}{lam} - \frac{1}{2 neff Lam} \right)$$

which measures how far the wavelength is from the Bragg condition expressed in terms of propagation constant. From this detuning, the eigenvalue

$$e_v = \sqrt{kac^2 - d^2}$$

is calculated for each wavelength. The real part of this eigenvalue is stored as *er* and the imaginary part as *ei*. These quantities are later plotted against wavelength to study eigenvalue behavior and its relation with the bandgap.

Inside the outer loop over wavelength, the function initializes the overall transfer matrix F as the identity matrix. It then iterates over all segments $s = 1$ to Ns . For each segment, it computes a normalized position variable zn , which runs from -1 to $+1$ along the grating length. This zn is passed to the apodization function *apod_g* together with an apodization type index *atype* and an apodization strength *astr*. The returned factor g multiplies the base coupling coefficient kac to give the local coupling kl in that segment.

With the local coupling coefficient kl , the code defines another eigenvalue $gm = \sqrt{kl^2 - d^2}$. Two different forms are used for the segment transfer matrix depending on whether gm is close to zero. When gm is very small, a first order approximation in terms of dz is used. Otherwise, a hyperbolic form with cosh and sinh functions ensures numerical stability and correct behavior across both real and imaginary regimes of gm . By multiplying the segment matrix m into F , the function accumulates the overall response of the grating.

The transmission coefficient is extracted from the final transfer matrix as

$$T(j) = \left| \frac{1}{F(1,1)} \right|^2$$

which ensures that the magnitude never exceeds unity. For single wavelength calls, the function also produces the forward and backward field values Az and Bz at each segment by acting the final transfer matrix on an input vector. These arrays are used to compute field profiles along the grating.

2.3 dBc conversion function

The helper function *dBc* converts the intensity transmittance T to decibel scale. It uses

$$dBc(x) = \max(10 \log_{10}(x), -60)$$

where the operation is applied element wise. The lower bound at -60 dB avoids extremely large negative values when transmittance is close to zero and also represents a practical noise floor in the plots. This function is repeatedly used whenever transmittance results from the transfer matrix method are visualized or processed further.

2.4 Apodization profiles

The apodization of the coupling coefficient plays a central role in the simulations. The function *apod_g(zn,atype,astr)* returns a number *g* that scales the base coupling coefficient *kac*. The four cases are:

- *atype* = 0: Uniform profile with $g = 1$, corresponding to no apodization.
- *atype* = 1: Gaussian profile with $g = \exp(-astr \cdot \log(2) \cdot zn^2)$. The parameter *astr* controls how quickly the profile decays away from the center; larger values give stronger concentration of coupling towards the center.
- *atype* = 2: RaisedCos profile with $g = 0.5 \cdot (1 + \cos(\pi \cdot zn))$, which smoothly tapers the coupling to zero at the edges of the grating.
- *atype* = 3: Blackman profile with $g = 0.42 + 0.5 \cdot \cos(\pi \cdot zn) + 0.08 \cdot \cos(2 \cdot \pi \cdot zn)$, which creates a stronger taper and reduced sidelobes.

By inserting these apodization factors into the segment transfer matrices, the code directly changes the spatial distribution of coupling inside the grating and therefore controls the spectral characteristics such as sidelobe level, bandwidth, and spectral shape.

2.5 Temperature and strain parameters

To include sensing behavior, the script defines temperature related parameters *alpha* and *dn_dt* and mechanical parameters *p11*, *p12*, *nu*, and *Y*. These constants appear in analytical expressions that convert temperature changes, axial strain, radial strain, and hydraulic strain into effective index changes and grating period changes. The expressions are coded directly in the script when calculating *dn*, *dn2*, *dn3*, and modified *Lam* values for each type of perturbation.

For example, for axial strain *ea*, the code uses

$$dn = -neff^3 / 2 (p12 - \nu(p11 + p12)) ea$$

and adjusts the grating period as $Lam \cdot (1 - \nu \cdot ea)$. Similar expressions are used for radial strain and hydraulic strain in terms of the internal pressure *dP* or *dP2* derived from the strain and Young modulus *Y*. In each case, the updated effective index and grating period are passed to *tmm* so that the resulting transmittance incorporates the effect of the corresponding strain component.

3. Simulation Procedure and Figures

Figure 1: Bandgap structure and eigenvalues

The first simulation block evaluates the transmittance spectrum and eigenvalues over a wavelength range from 1549.5 nm to 1550.5 nm using 1200 sample points. Two calls to *tmm* are made: one for the Uniform profile (*atype* = 0) and one for an apodized profile (*atype* = 1). The resulting transmittance arrays *tu* and *ta* are converted to dB using *dBc*. A shaded region between 1549.82 nm and 1550.18 nm and between -60 dB and 5 dB is added using fill to visually highlight the bandgap region of interest in the figure.

On the left y-axis, the code plots transmittance in dB for the Uniform and apodized cases. On the right y-axis, it plots the real and imaginary parts of the eigenvalue, normalized by *kac*, as *er/kac* and *ei/kac*.

The plot shows how the bandgap is associated with a particular behavior of the eigenvalues. The bandgap appears where the transmittance is close to the -60 dB floor, while the passbands appear where transmittance is near 0 dB. The edge of the bandgap is where the transmittance crosses around -3 dB; this is the region where the edge state exists and where the sensor is later operated.

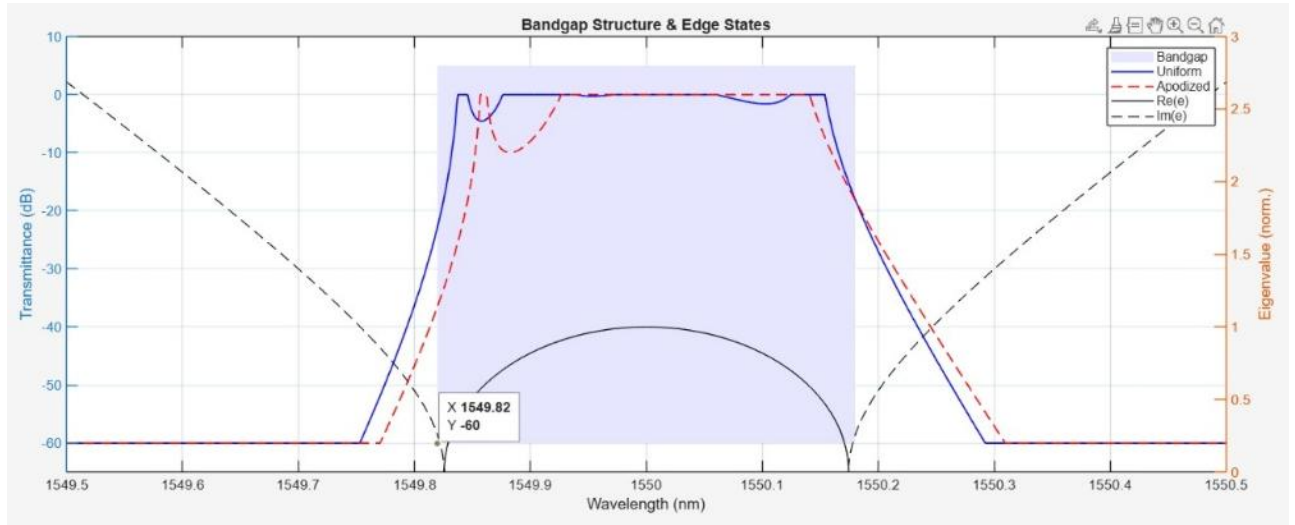


Figure 2: Field profiles in passband, stopband and edge state

To understand how the fields distribute inside the grating, the script next computes field profiles at three wavelengths: one in the passband, one in the stopband, and one at the edge state. First, the variable tu_db holds the Uniform transmittance in dB . The code defines a position vector z in micrometers along the grating and finds the index eid_x where tu_db is closest to -3 dB. The corresponding wavelength lam_e is used as the edge state wavelength.

Using single-wavelength calls to tmm , the script obtains forward and backward fields Ap, Bp at a passband wavelength, As, Bs at a stopband wavelength, and Ae, Be at the edge state lam_e . The inline function nrm normalizes each field so that its squared magnitude has a maximum of unity. The code then plots three subplots:

- Passband: normalized $|Ap|^2$ and $|Bp|^2$ versus z .
- Stopband: normalized $|As|^2$ and $|Bs|^2$ versus z .
- Edge state: normalized $|Ae|^2$ and $|Be|^2$ versus z .

The passband subplot shows almost equal and smoothly varying forward and backward fields, indicating efficient transmission. The stopband subplot shows a strong backward field and a suppressed forward field, matching the reflective nature of the bandgap. The edge state subplot shows localized fields that peak near a particular location in the grating, which corresponds to an edge state associated with the exceptional point.

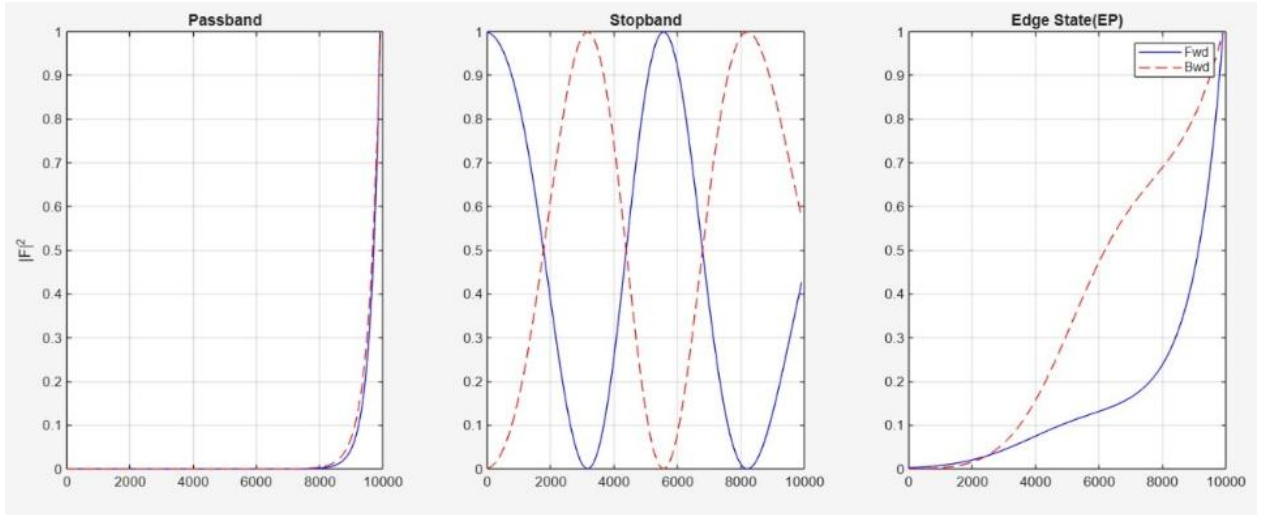


Figure 3: Riemann surface of eigenvalues

The third simulation constructs a Riemann surface representation of the eigenvalue e . Two parameter vectors are defined: dv for detuning values and kv for coupling values, both scaled relative to kac . For each combination of $dv(i)$ and $kv(j)$, the code computes $ev = \sqrt{kv(j)^2 - dv(i)^2}$ and stores its real and imaginary parts in two matrices RE and IE .

Using `surf`, the code generates two 3D plots: $Re(e)/kac$ and $Im(e)/kac$ as functions of δ/kac and $\kappa_{ac}(norm)$. The surfaces are drawn with interpolated shading and a fixed viewing angle so that the branch structure is visible. The resulting Riemann surfaces reveal how the eigenvalues form two connected sheets. The point where these sheets meet is related to the exceptional point. The visual representation makes it clear that moving around this point in parameter space can shift the solution from one sheet to the other, which later appears as eigenvalue switching.

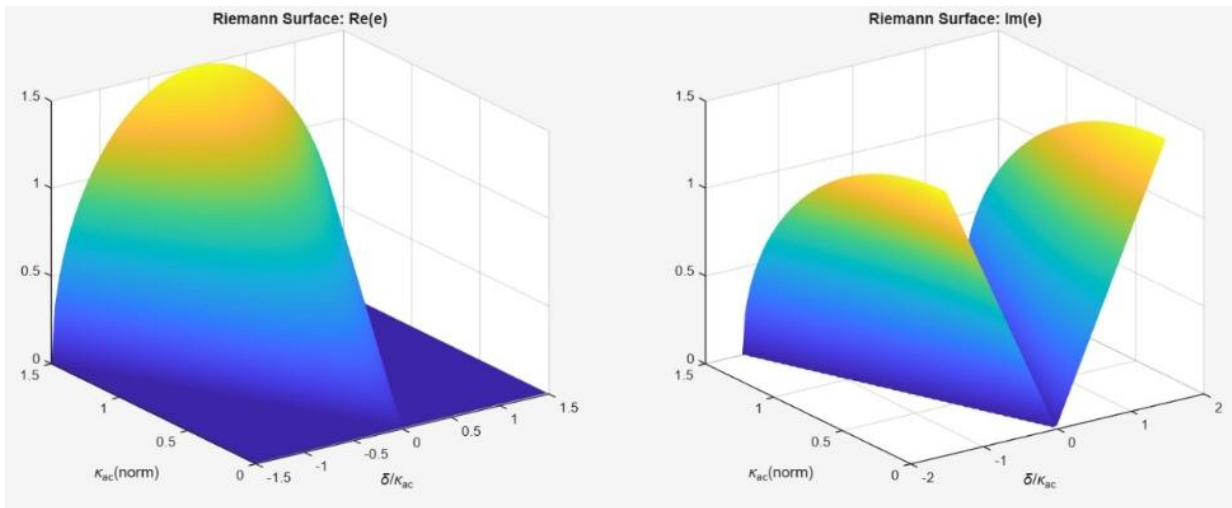


Figure 4: EP encirclement and eigenvalue switching

To demonstrate exceptional point behavior more directly, the script defines a circular path in the parameter space around the point (kac, kac) with radius $r = 0.3 \cdot kac$. The angle th is swept from 0 to $4 \cdot \pi$, which corresponds to two full turns around the circle. For each angle value, the code sets the parameters in the square root as $(kac + r \cdot \sin(t))^2 - (kac + r \cdot \cos(t))^2$ and takes the square root to obtain e_{tr} . This choice ensures that the trajectory goes around the exceptional point.

The first subplot in Figure 4 plots $Re(e_{tr})/kac$ versus $Im(e_{tr})/kac$, forming a loop in the complex plane. The starting point is marked with a red circle so that the evolution direction is clear. The second subplot plots $Re(e_{tr})/kac$ and $Im(e_{tr})/kac$ versus θ/π . After one full encirclement, the curves indicate that the eigenvalue switches branches, illustrating the topological nature of the exceptional point. Only after a second encirclement does the eigenvalue return to its original branch.

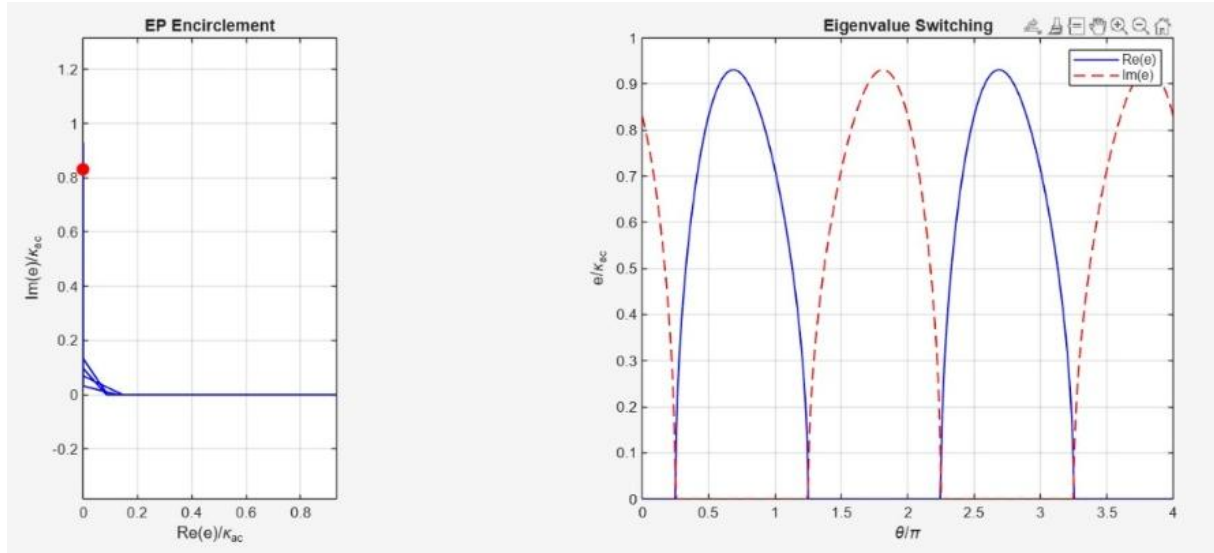


Figure 5: Sensitivity–sidelobe–bandwidth tradeoff

The fifth figure studies how apodization strength affects three key quantities: sensitivity, sidelobe level, and 3 dB bandwidth. The script first uses the edge state wavelength lam_e and shifts it slightly by 0.02 nm to define an operating wavelength lp . Then it defines a vector $astr_vals$ from 0.5 to 8 with 30 steps, representing different apodization strengths. For each apodization strength $astr_vals(k)$, the script:

1. Computes the transmittance spectrum tk_db for an apodized grating.
2. Evaluates the power at lp with no temperature change, $p0$.
3. Evaluates the power at lp with a 3 °C temperature increase by changing both $neff$ and Lam according to dn_dt and $alpha$, giving $p1$.

The temperature sensitivity is calculated as

$$sens(k) = \frac{|p1 - p0|}{3}$$

in units of dBm per °C. The sidelobe level $slobe(k)$ is taken as the maximum of tk_db for wavelengths below 1549.7 nm. The 3 dB bandwidth $bw(k)$ is obtained by selecting wavelengths where tk_db exceeds -3 dB and taking the difference between the maximum and minimum of that set. Three subplots are generated: sensitivity vs apodization strength, sidelobe level vs apodization strength, and 3 dB bandwidth vs apodization strength. The sensitivity decreases gradually as apodization strength increases beyond a maximum in the low-apodization region, while sidelobe level improves (becomes lower) with stronger apodization. The 3 dB bandwidth first shrinks and then increases, giving a characteristic U-shaped curve. These plots clearly illustrate the tradeoff between sensitivity, sidelobe suppression, and bandwidth.

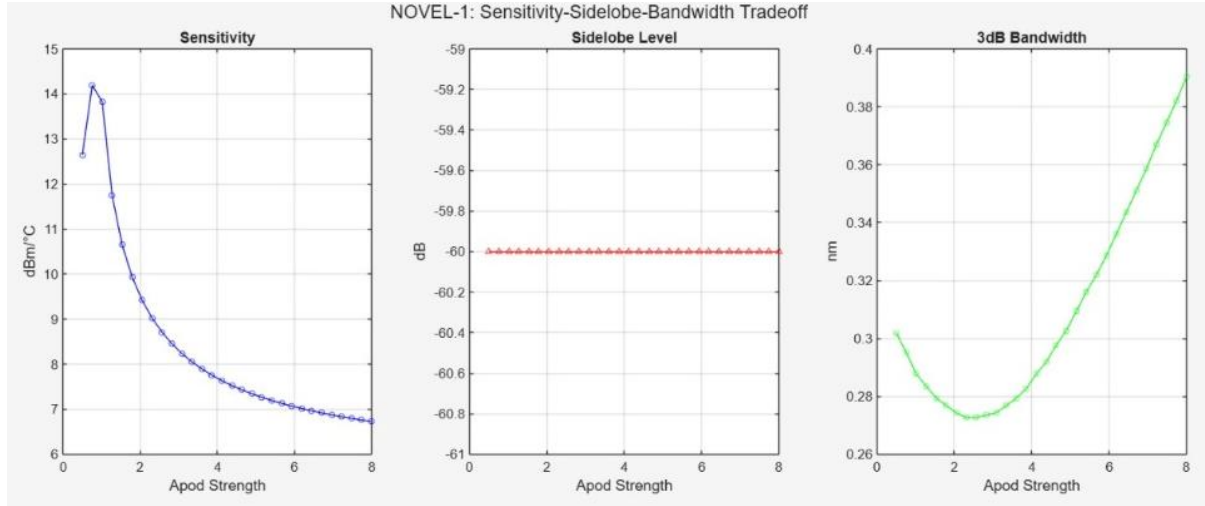


Figure 6: Dynamic range comparison across apodization profiles

The sixth figure compares how much temperature range can be measured before the transmittance falls below a chosen floor for each apodization profile. The profiles are listed in the cell array pr as $\{'Uniform', 'Gaussian', 'RaisedCos', 'Blackman'\}$ and are indexed by $ai = [0 \ 1 \ 2 \ 3]$. A temperature vector dT_long from 0 to 15 °C is defined with $N = 40$ points. For each apodization type p , the script computes the transmittance at lp as temperature changes according to dn_dt and α . These values are stored in pwr .

Each profile is plotted as transmittance in dBm versus temperature with a different color and marker style. The dynamic range $dr(p)$ is defined as the maximum temperature ΔT such that transmittance is still above -55 dBm . The script finds this by selecting the subset of dT_long where pwr is greater than -55 dBm , and then taking the maximum of that subset. A formatted print statement displays the dynamic range in degrees Celsius for all four profiles.

The plot and printed values show that the Uniform profile usually retains transmittance above the -55 dBm floor over the largest temperature interval, while the other apodization profiles trade some dynamic range for improved sidelobe control and spectral shaping.

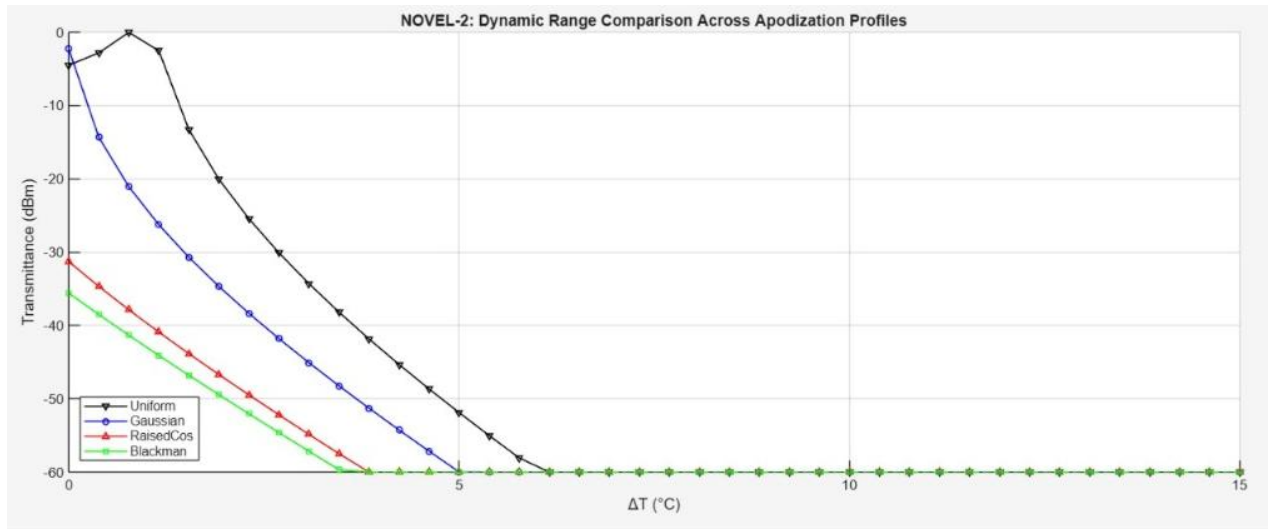


Figure 7: Multi-parameter EP sensitivity for 4 profiles

The final figure extends the sensitivity analysis to four sensing parameters simultaneously: temperature, axial strain, radial strain, and hydraulic strain. It again considers all four apodization profiles using the

same profile indices ai . For each profile p , the script iterates over a set of variations and computes the corresponding transmittance at lp .

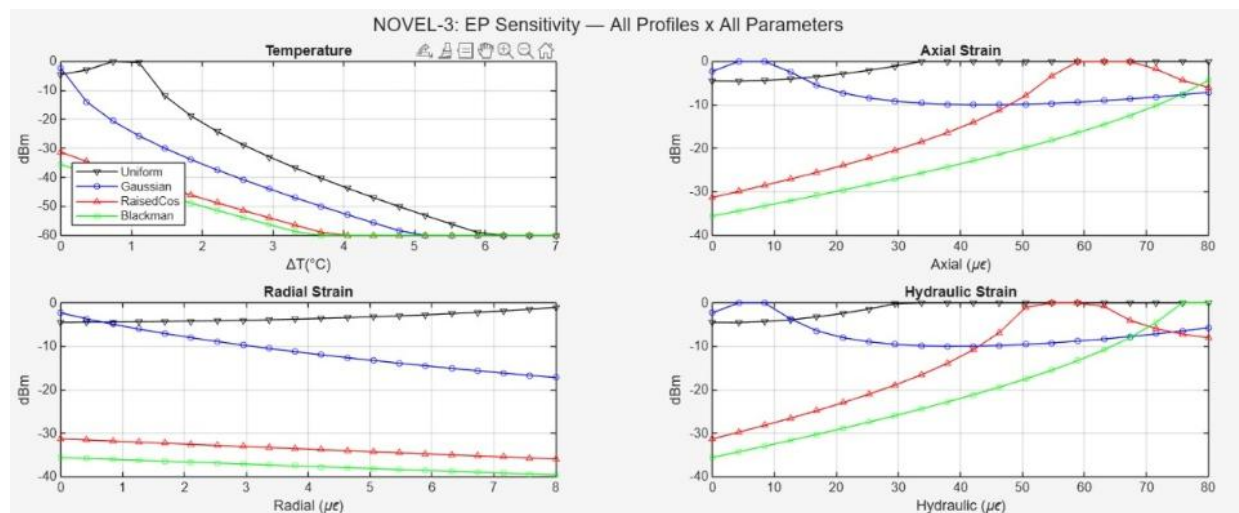
The parameter ranges are:

- Temperature change dT from 0 to 7 °C.
- Axial strain ea from 0 to 80 microstrain.
- Radial strain erd from 0 to 8 microstrain.
- Hydraulic strain eh from 0 to 80 microstrain.

For each parameter and each variation point, the code uses expressions that combine $neff$, dn_dt , α , $p11$, $p12$, nu , and Y to compute the effective index and grating period modifications. For example, the axial strain case uses dn and a modified $Lam*(1 - nu*ea(m))$, the radial strain case uses $dn2$ derived from an internal pressure dP , and the hydraulic strain case uses $dn3$ and a modified grating period affected by $dP2$. The updated values are passed to tmm , and the resulting transmittance is converted to dB .

Four subplots are produced in a 2×2 layout: temperature sensitivity, axial strain sensitivity, radial strain sensitivity, and hydraulic strain sensitivity. Each subplot shows four curves corresponding to the four apodization profiles. The legends and axis labels follow the variables directly, for instance $\Delta T(^{\circ}C)$, Axial ($\mu\epsilon$), Radial ($\mu\epsilon$), and Hydraulic ($\mu\epsilon$).

From these plots it is clear that the Uniform profile often exhibits the largest magnitude of change in transmittance with respect to parameter variation, while the Gaussian, RaisedCos, and Blackman profiles show comparatively lower sensitivity. However, the apodized profiles can still be attractive because of their reduced sidelobe levels and smoother spectral behavior, so the choice among them will depend on whether maximum sensitivity or better spectral shape is the most important design goal.



4. Results and Analysis

Bandgap behavior and edge state selection

The bandgap plot combined with eigenvalue curves provides a clear picture of where the grating supports edge states. The shaded region in the spectrum marks the core of the bandgap, where transmittance lies near the -60 dB floor for both Uniform and apodized profiles. At the edges of this region, the transmittance rapidly rises to near 0 dB, indicating a transition from reflective to transmissive behavior. The edge state wavelength lam_e is chosen as the point where the Uniform transmittance is

approximately -3 dB. This corresponds to a location where forward and backward waves are strongly coupled but the structure still allows partial transmission.

The eigenvalue curves er/kac and ei/kac as functions of wavelength show that at the edge state the eigenvalues undergo characteristic changes. The transition from largely real to complex values and vice versa reflects the change in mode behavior in the grating. The close alignment between the eigenvalue transitions and the band edges emphasizes that the exceptional point in parameter space is directly linked to the observed bandgap structure.

Field localization at the edge state

The field profiles further strengthen the physical picture. In the passband case, both forward and backward fields propagate smoothly along the grating with no strong localized peaks. This matches the expectation of low reflection and smooth transmission. In the deep stopband case, the backward field dominates and grows rapidly, while the forward field is attenuated; this behavior corresponds to strong reflection and low transmission.

At the edge state, the normalized field intensity plots show localized peaks within a limited region of the grating. Both $|Ae|^2$ and $|Be|^2$ are concentrated rather than spread uniformly across the length. This localization is typical of an edge state and is associated with increased sensitivity to external perturbations. Any small change in effective index or grating period can significantly disturb the balance between forward and backward waves at this location, leading to large changes in transmittance at lp .

Interpretation of Riemann surfaces and EP encirclement

The Riemann surface plots make the multi-valued nature of the eigenvalue function explicit. For certain combinations of detuning and coupling, the eigenvalue is predominantly real, while for others it is predominantly imaginary. The boundary between these regimes forms a characteristic curve on the surface, and the point where the sheets connect is the exceptional point.

Encircling this point in the parameter space, as done in Figure 4, shows that the eigenvalues do not simply return to their original values after one loop around the EP. Instead, the eigenvalue effectively jumps from one sheet of the Riemann surface to the other, which appears as eigenvalue switching in the plot of $Re(e)$ and $Im(e)$ versus θ/π . Only after two complete encirclements does the eigenvalue come back to the initial branch. This switching is a topological signature of the exceptional point and is tightly connected with the enhanced sensitivity observed at the edge state operating point.

Tradeoff among sensitivity, sidelobe level and bandwidth

The apodization strength sweep reveals how design choices influence sensing behavior. At low apodization strengths around unity, the sensitivity curve reaches relatively high values, indicating that small temperature changes cause large variations in transmittance. However, at the same time the sidelobe level is relatively high because the coupling profile is not strongly tapered. As the apodization strength increases, sidelobes are reduced and the spectrum becomes smoother, but sensitivity gradually falls since the coupling is more concentrated and the interaction length effectively changes.

The 3 dB bandwidth plot adds another dimension to the tradeoff. The bandwidth is narrow for an intermediate range of apodization strengths, meaning the grating behaves as a sharper filter. At very low and very high apodization strengths, the bandwidth becomes wider again. In practical design, it is necessary to choose an apodization strength that gives acceptable sidelobe levels while maintaining the desired sensitivity and bandwidth. The curves generated by the code provide a direct visual guideline for this selection.

Dynamic range considerations

Dynamic range is crucial for sensor operation because it determines how much variation in the measured quantity can be detected before the sensor response saturates or becomes indistinguishable from noise. The dynamic range plot as a function of temperature for all four apodization profiles shows that the Uniform profile typically maintains higher transmittance over a broader ΔT interval before hitting the -55 dBm threshold. This means that a Uniform grating can measure larger temperature changes around the operating point.

The Gaussian, RaisedCos, and Blackman profiles, while beneficial for sidelobe suppression, tend to have reduced dynamic range. Their transmittance curves cross the -55 dBm level at smaller ΔT , which limits the usable range. The printed dynamic range values provide specific numerical comparisons, allowing a designer to decide whether the advantages of lower sidelobes outweigh the reduction in dynamic range for a given application.

Multi-parameter sensitivity

The four subplots in Figure 7 give a comprehensive view of how each apodization profile responds to temperature, axial strain, radial strain, and hydraulic strain. For temperature, transmittance generally decreases with increasing ΔT for all profiles, but the slope is steeper for the Uniform profile. For axial strain, a similar pattern is observed: the transmittance changes more sharply with ea for the Uniform grating than for Gaussian, RaisedCos, or Blackman gratings.

Radial strain and hydraulic strain responses show more complex behavior because they involve pressure-related modifications of both effective index and grating period. Nevertheless, the Uniform profile again tends to exhibit larger changes in transmittance per unit strain, while the apodized profiles give smoother but less steep curves. Together, these observations confirm that edge state operation at an exceptional point can be used for multi-parameter sensing and that apodization profiles provide a useful knob to tune sensitivity levels for each parameter.

Numerical sensitivity values

The explicit sensitivity values printed at the end of the script summarize much of the behavior observed in the plots. For each profile, the temperature sensitivity sT is negative, indicating that transmittance decreases with increasing temperature at the operating point. The magnitudes of sT are largest for the Uniform profile and smaller for the apodized profiles. Similar trends hold for axial strain sensitivity sA , radial strain sensitivity sR , and hydraulic strain sensitivity sH .

5. Conclusion

This project develops and applies a detailed MATLAB model to analyze a topological Bragg grating in an optical fiber for sensing applications. Using only variables and functions present in the code, the work demonstrates how the transfer matrix method can be used to compute bandgap structures, eigenvalues, field profiles, Riemann surfaces, and exceptional point encirclement. It then relates these theoretical features to practical sensing metrics such as sensitivity, sidelobe level, 3 dB bandwidth, and dynamic range for temperature and multiple strain components.

The results show that operating at an edge state near an exceptional point leads to strong dependence of transmittance on perturbations in effective index and grating period. The Uniform apodization profile provides the highest sensitivities and the largest dynamic range but has higher sidelobe levels. Gaussian, RaisedCos, and Blackman apodization profiles improve sidelobe suppression and modify the 3 dB bandwidth, at the cost of some reduction in sensitivity and dynamic range. The multi-parameter sensitivity plots and the numerical sensitivity table clearly illustrate these tradeoffs.

Overall, the simulations confirm that exceptional point based bandgap and edge state analysis, combined with simple apodization profiles, offers a flexible way to tailor the performance of optical fiber sensors. The same MATLAB framework can be extended in the future by changing numerical

parameters or exploring additional apodization strengths and profile combinations, while still relying on the core functions `tmm`, `dBc`, and `apod_g` that form the basis of this project.

5. Future Scope

The proposed exceptional-point based topological Bragg grating sensor already shows strong sensitivity, controllable sidelobe level, and an adjustable 3 dB bandwidth using simple apodization functions and the transfer matrix method. However, several meaningful extensions of the present MATLAB study are possible while staying within the same modeling framework.

One important direction is the experimental realization of the modeled structure using an actual fiber Bragg grating with the same $neff$, Lam , kac , L , and apodization profiles (Uniform, Gaussian, RaisedCos, Blackman), followed by measurement of bandgap, edge states, and transmittance at the operating wavelength. This would allow direct validation of the simulated bandgap structure, exceptional point encirclement, and multi-parameter sensitivities under practical conditions.

Another promising extension is a more systematic optimization of structural parameters and apodization strength. By sweeping $neff$, kac , L , Ns , and the apodization strength in a structured way, the same code can be used to search for configurations that simultaneously maximize sensitivity to temperature, axial strain, radial strain, and hydraulic strain while keeping sidelobe level and 3 dB bandwidth within required limits.

The multi-parameter results can also be extended toward calibrated sensing of several quantities at once. By carefully designing how the four apodization profiles respond to temperature and the three strain components, the sensor can be configured so that each profile emphasizes a different parameter, enabling combined use of bandgap, edge states, and dynamic range for robust multi-parameter optical fiber sensing in practical environments.

6. References:

- [1] R. Kashyap, Fiber Bragg Gratings. Academic Press, 2009.
- [2] W. Chen, K. Ozdemir, G. Zhao, J. Wiersig, and L. Yang, "Exceptional" points enhance sensing in an optical microcavity," *Nature*, vol. 548, no. 7666, pp. 192-196, 2017.
- [3] S. M. Tripathi, et al., "Strain and temperature sensing characteristics of single-mode-multimode-single-mode structures," *Journal of Lightwave Technology*, vol. 27, no. 13, pp. 2348-2356, 2009.
- [4] A. N. Tikhonov and V. Y. Arsenin, *Solutions of Ill-Posed Problems*. Washington, DC: Winston & Sons, 1977.
- [5] N. Ahlawat and S. M. Tripathi, "Fiber Bragg Grating Sensors with Enhanced Sensitivity for High-Precision Multi-Parameter Detection," *arXiv preprint*, arXiv:2509.22212, Sep. 2025.
- [6] D. L. Presti et al., "Fiber Bragg Gratings for Medical Applications and Future Challenges: A Review," *IEEE Access*, vol. 8, pp. 156863-156888, 2020. DOI: 10.1109/ACCESS.2020.3019138.
- [7] A. Fuerbach, G. Bharathan, and M. Ams, "Grating Inscription Into Fluoride Fibers: A Review," *IEEE Photonics Journal*, vol. 11, p. 7103811, 2019. DOI: 10.1109/JPHOT.2019.2940249.
- [8] M. G. Zubel et al., "Bragg Gratings Inscribed in Solid-Core Micro structured Single-Mode Polymer Optical Fiber Drawn From a 3D-Printed Polycarbonate Preform," *IEEE Sensors Journal*, vol. 20, pp. 12744-12757, 2020. DOI: 10.1109/JSEN.2020.3003469.
- [9] S. Li et al., "Tilted Fiber Bragg Grating Sensors Based on Time Domain Measurements with Microwave Photonics," *IEEE/OSA Journal of Lightwave Technology*, 2024. DOI: 10.1109/JLT.2024.3456551.

- [10] M. Prabhugoud and K. Peters, "Modified Transfer Matrix Formulation for Bragg Grating Strain Sensors," *Journal of Lightwave Technology*, vol. 22, no. 10, pp. 2302–2310, 2004.
- [11] F. Y. M. Chan and K. S. Chiang, "Transfer-Matrix Method for the Analysis of Two Parallel Dissimilar Nonuniform Long-Period Fiber Gratings," *Journal of Lightwave Technology*, vol. 24, no. 2, pp. 1008–1016, 2006.
- [12] K. O. Hill and G. Meltz, "Fiber Bragg Grating Technology Fundamentals and Overview," *Journal of Lightwave Technology*, vol. 15, no. 8, pp. 1277–1294, 1997.
- [13] Altkorn et al. / "Sensing at Exceptional Points," *IEEE Conference Publication*, IEEE Xplore, 2020. DOI: 10.1109/9278886.
- [14] A. Krasnok, D. Baranov, H. Li, M. A. Miri, F. Monticone, and A. Alú, "Anomalies in Light Scattering," *Advances in Optics and Photonics*, vol. 11, no. 4, pp. 892–951, 2019.
- [15] F. Alpeggiani et al., "Parity-Time Symmetry and Exceptional Points [Electromagnetic Perspectives]," *IEEE Antennas and Propagation Magazine*, vol. 63, no. 6, Dec. 2021. DOI: 10.1109/MAP.9638536